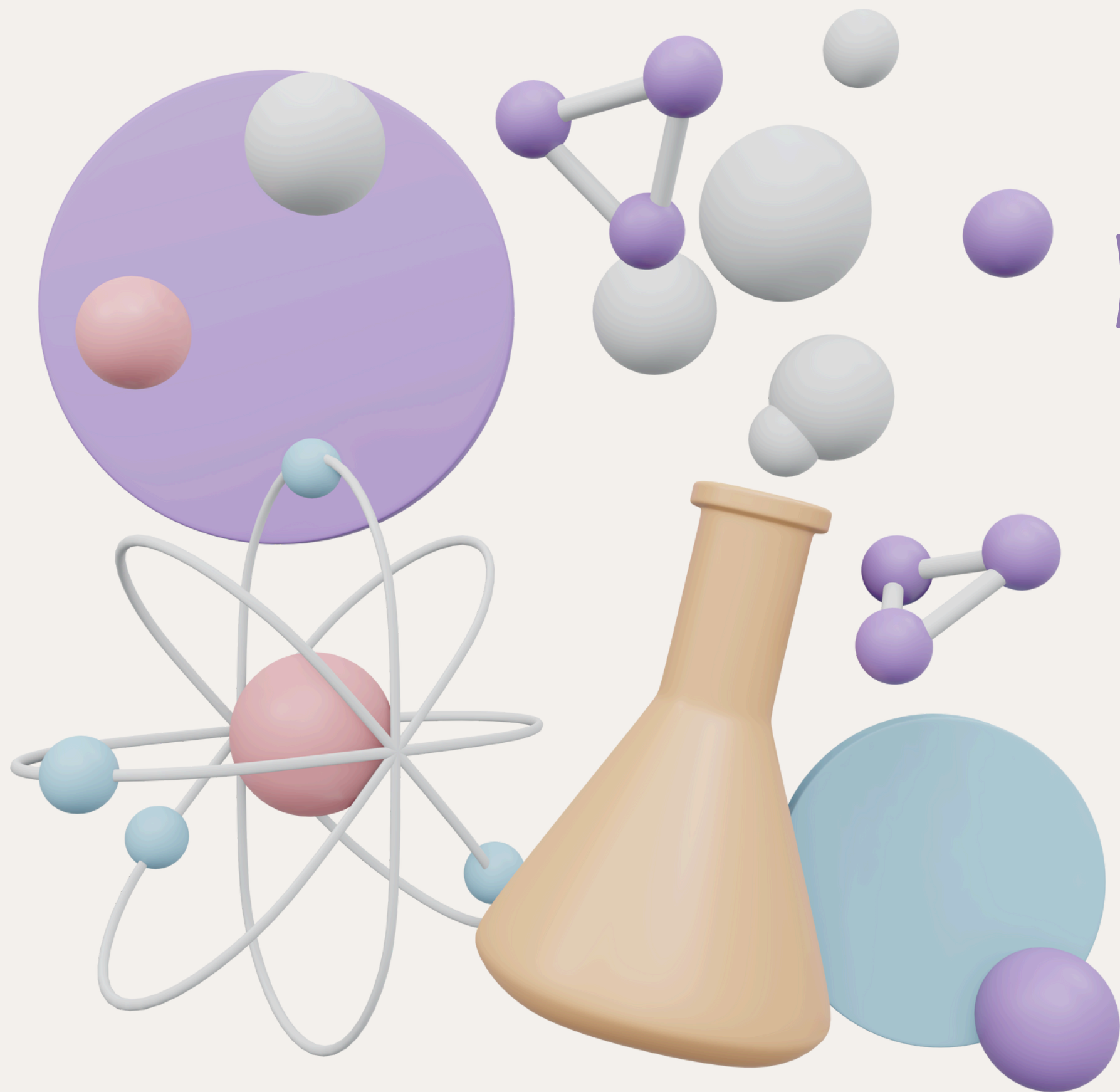




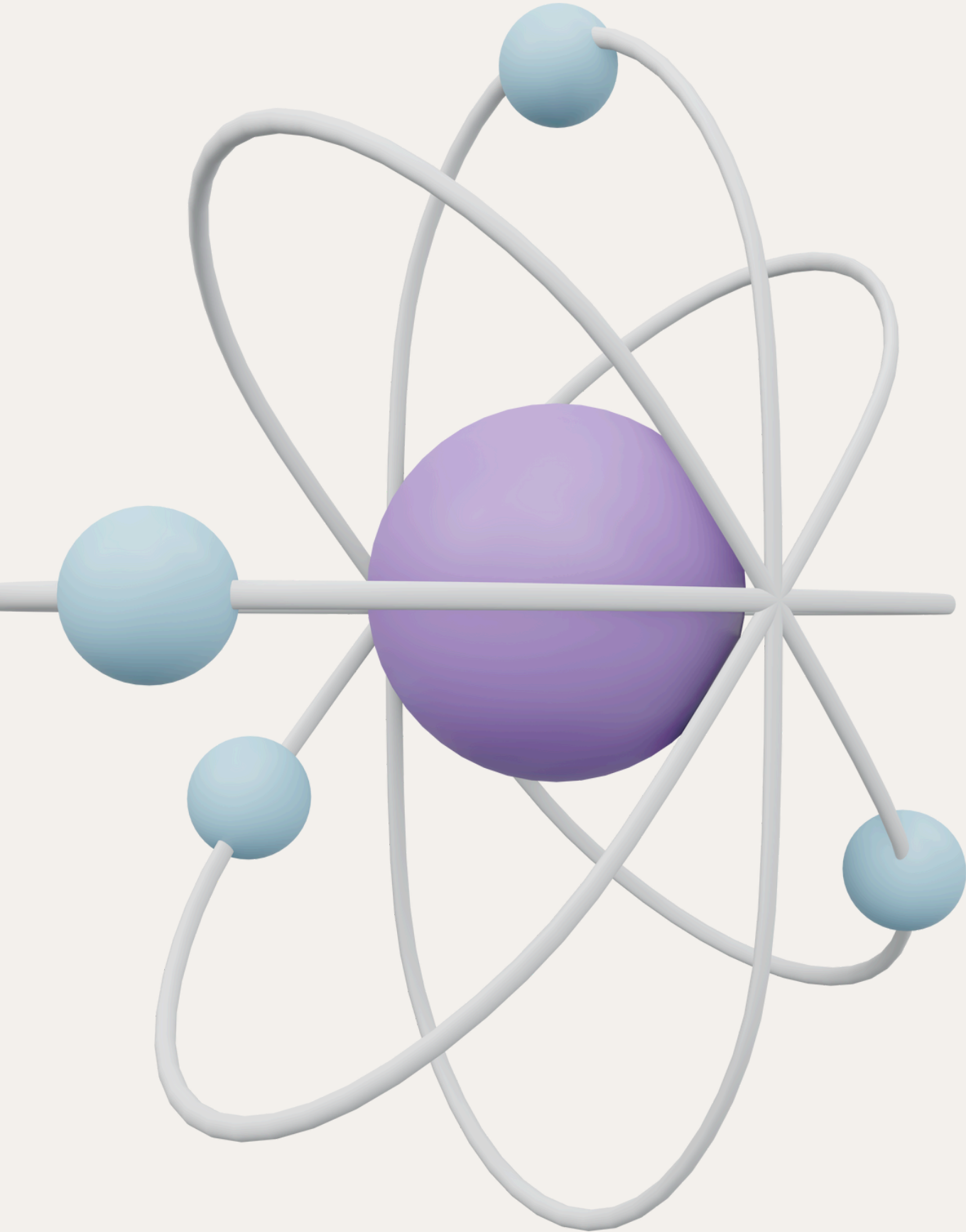
BUILDERS, FILLERS, AND ADDITIVES: ROLES AND INNOVATIONS

Group 3

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WHY ARE BUILDERS CRUCIAL IN CLEANING PERFORMANCE?

Role	Key Action	Why are they crucial
WATER SOFTENING	remove the hard water mineral (Ca ²⁺ & Mg ²⁺)	Boost Surfactant Power: Deactivate the ions into soap scum (precipitate) to allow the cleaning agents to work effectively.
PH CONTROL	Provide alkalinity (high pH) and maintain cleaning power (buffering)	Helps in breaking down fatty and oily soils and ensures consistent cleaning throughout the wash cycle.





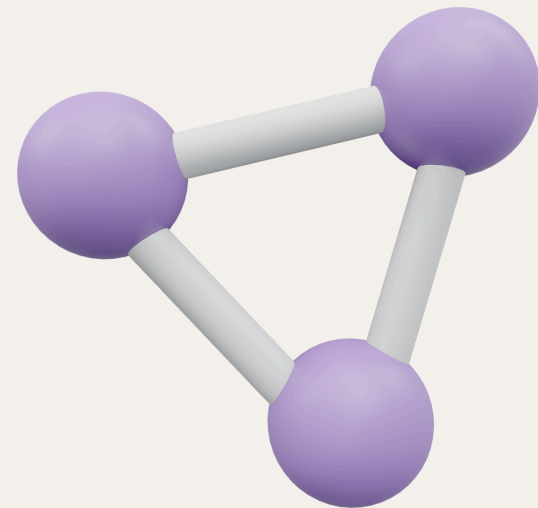
ARE OPTICAL BRIGHTENERS SAFE?

Function: Absorb invisible UV light and re-emit visible blue light

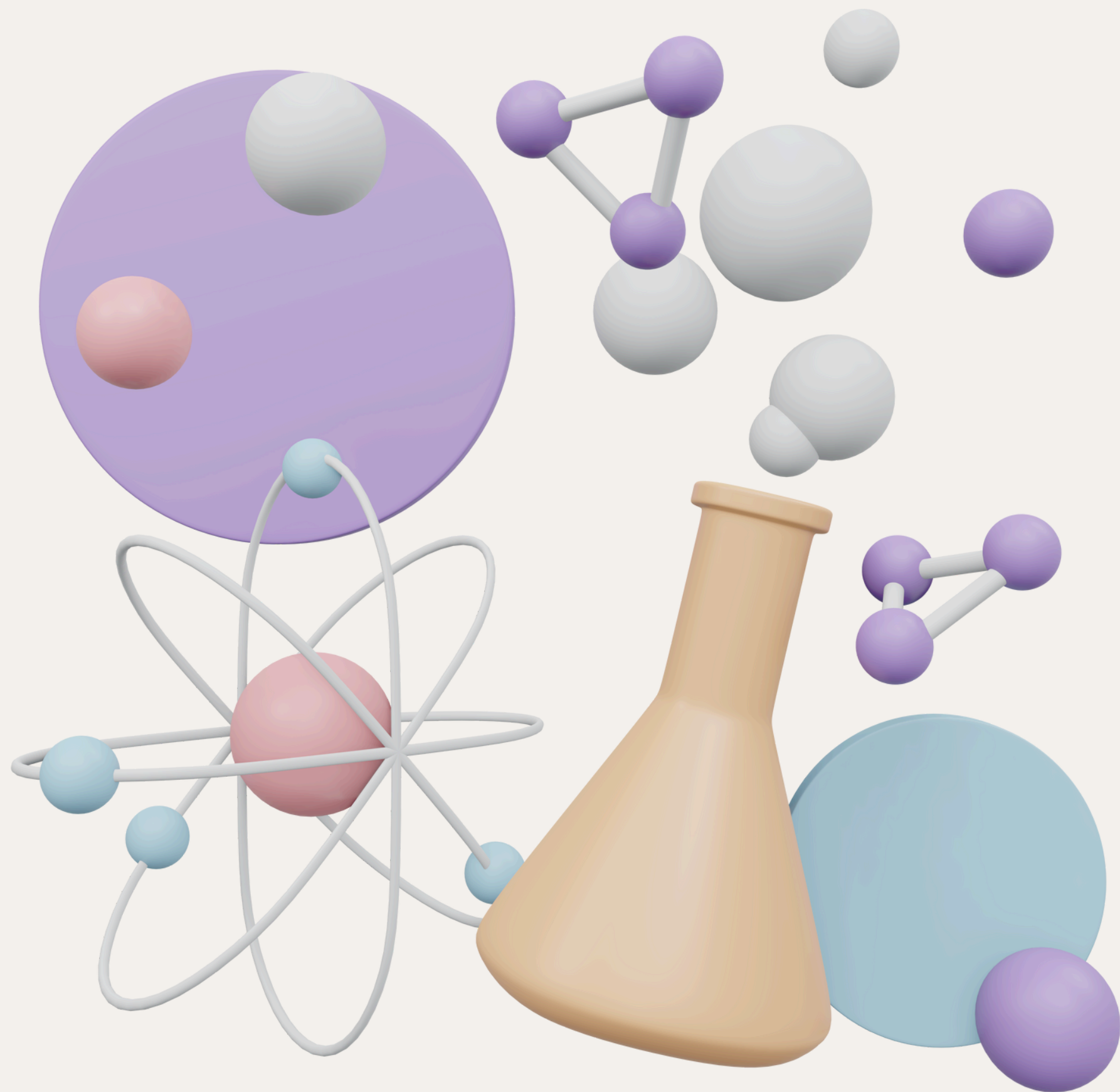
Result: Masks yellowing, making fabrics appear whiter and brighter (don't actually clean)

Safety note: Generally low risk, but environmental concerns (non-biodegradable) and potential for skin sensitivity exist.

ADVANCES OF ENZYME TECHNOLOGY



Enzyme Type	PROTEASE	LIPASE	AMYLASE
Target Stain	Protein (Blood, Grass)	Fat/Oil (Grease, Butter)	Starch (Pasta, Gravy)
Innovation Focus	Engineered Stability: Stable in liquid detergents and high-pH environments	Low-Temp Efficiency: Work powerfully in cold water, saving energy	Multi-Enzyme Systems: Blends of enzymes attack a wider range of complex stains synergistically.



**THANK YOU
SO MUCH!**